

## SUMMARY

Computational scientist with a strong foundation in method development and scientific software engineering. Specialized in machine learning and nonadiabatic molecular dynamics: learning of potential energy surfaces, nonadiabatic couplings, spin-orbit couplings and dipole moments, a flexible surface hopping framework and visualization tools.

## TECHNICAL SKILLS

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|---------------------------|----------------------------------------------------------------------------------|
| <b>Programming</b>        | Python, C/C++, Fortran, Bash/Shell                                               |
| <b>Scientific Python</b>  | NumPy, SciPy, Pandas, Matplotlib, Scikit-learn, PyTorch                          |
| <b>Method development</b> | MLatom (Python), Newton-X (C++/Fortran)                                          |
| <b>Quantum chemistry</b>  | Columbus, Molcas, Turbomole, BAGEL, Molpro, Psi4, PySCF, Gaussian                |
| <b>Tools</b>              | Linux/UNIX, Git/GitHub, vim/VSCode, L <sup>A</sup> T <sub>E</sub> X, Claude Code |
| <b>Web development</b>    | HTML, CSS, JavaScript, PHP, SQL, SharePoint (UniCredit Bank, 2020–present)       |
| <b>Languages</b>          | Czech (native), English (C1), German (A2), Japanese (A1)                         |

## EDUCATION

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|-------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>PhD in Modeling of Chemical Nanostructures and Biostructures</b>                                               | 10/2022 – present |
| <i>Faculty of Science, Charles University</i>                                                                     |                   |
| Thesis: “Excited state molecular dynamics with non-adiabatic and spin-orbit effects assisted by machine learning” |                   |
| Supervisor: J. Pittner — Dept. of Theoretical Chemistry, Heyrovský Institute, CAS (since 12/2018)                 |                   |
| <b>MSc in Physical Chemistry</b> (summa cum laude)                                                                | 10/2020 – 9/2022  |
| <i>Faculty of Science, Charles University</i>                                                                     |                   |
| Thesis: “Non-adiabatic molecular dynamics of photochemical processes accelerated by machine learning”             |                   |
| <b>BSc in Biochemistry</b>                                                                                        | 10/2017 – 9/2020  |
| <i>Faculty of Science, Charles University</i>                                                                     |                   |
| Thesis: “Molecular modeling of the interaction between cytochrome P450s and their substrates/products”            |                   |

## PUBLICATIONS

- [1] Martinka J., Sit M. K., Dral P. O. and Pittner J., “Toward Machine Learning Enhancement of Accelerated Surface Hopping with Scaled Spin-Orbit Couplings”, *Under review in JPCA*.
- [2] Martinka J., Martyka M., Pittner J. and Dral P. O., “Flexible Framework for Surface Hopping: From Hybrid Schemes for Machine Learning to Benchmarkable Nonadiabatic Dynamics”, *Journal of Chemical Theory and Computation*, **22**, 2467-2479 (2026).
- [3] Martinka J., Zhang L., Hou Y.-F., Martyka M., Pittner J., Barbatti M. and Dral P. O., “A Descriptor Is All You Need: Accurate Machine Learning of Nonadiabatic Coupling Vectors”, *Journal of Physical Chemistry Letters*, **16**, 11732–11744 (2025).
- [4] Martinka J., Pederzoli M., Barbatti M., Dral P. O. and Pittner J., “A Simple Approach to Rotationally Invariant Machine Learning of a Vector Quantity”, *Journal of Chemical Physics*, **161**, 174104 (2024).

## RESEARCH STAYS & FELLOWSHIPS

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|-----------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Barrande Fellowship with Prof. M. Barbatti</b> (Aix-Marseille Université, France)                                  | 9/2026 – 11/2026  |
| Extending Newton-X (Fortran) to intersystem crossing, enabling surface hopping simulations with spin-orbit couplings. |                   |
| <b>Research visit with Prof. H. Lischka</b> (Texas Tech University, Lubbock, TX, USA)                                 | 5/2026            |
| ML in nonadiabatic molecular dynamics within MLatom (Python).                                                         |                   |
| <b>Research visit with Prof. P. O. Dral</b> (Xiamen University, Xiamen, China)                                        | 10/2024 – 11/2024 |
| Implementation of fewest-switches surface hopping in MLatom (Python) and its combination with ML potentials.          |                   |
| <b>Research visit with Prof. M. Barbatti</b> (Aix-Marseille Université, France)                                       | 1/2024            |
| Machine learning and sign correction of nonadiabatic coupling vectors (Python).                                       |                   |

## GRANTS, AWARDS & SERVICE

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| <b>Grants &amp; funding</b> | Barrande Fellowship (2453), GAUK 2024 (project 6224), POINT 2024 (2024-1-194), Mobility Fund (2026-1-031), GAČR (project 23-06364S), AMULET project                                            |
| <b>Awards</b>               | ACS Best Poster Award, 20th CESTC (2025); 3× talk award, student conferences of the Heyrovský Institute (2023–2025); Merit scholarship for excellent academic performance (Charles University) |
| <b>Peer review</b>          | <i>J. Chem. Theory Comput.</i> (2025), <i>Nat. Commun.</i> (2026)                                                                                                                              |

## CONFERENCES (POSTER PRESENTATIONS)

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|-----------------------------------------------------------------------------------------|-------------------|
| <b>18th International Congress of Quantum Chemistry</b> (Berkeley, CA, USA)             | 31/5 – 6/6/2026   |
| <b>2nd Chemical Compound Space Conference</b> (Munich, Germany)                         | 10 – 13/2/2026    |
| <b>3rd International Symposium on ML in QCh</b> (Knoxville, TN, USA)                    | 5 – 7/10/2025     |
| <b>20th Central European Symposium on Theoretical Chemistry</b> (High Tatras, Slovakia) | 8 – 11/9/2025     |
| <b>13th Triennial Congress of the WATOC</b> (Oslo, Norway)                              | 22 – 27/6/2025    |
| <b>11th Triennial Congress of the ISTCP</b> (Qingdao, China)                            | 13 – 18/10/2024   |
| <b>WTC Mariapfarr – Machine Learning for Chemistry</b> (Bad Hofgastein, Austria)        | 27/2 – 1/3/2024   |
| <b>2nd International Symposium on ML in QCh</b> (Uppsala, Sweden)                       | 29/11 – 1/12/2023 |
| <b>17th International Congress of Quantum Chemistry</b> (Bratislava, Slovakia)          | 26/6 – 1/7/2023   |
| <b>ICQC Satellite Meeting on Strong Correlation in Molecules</b> (Znojmo, Czechia)      | 20 – 23/6/2023    |

## SUMMER SCHOOLS & WORKSHOPS

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| <b>Machine Learning for Chemistry Summer School</b> (KIT, Karlsruhe, Germany)                 | 9 – 13/9/2024   |
| Molecular descriptors, GNNs, ML potentials, synthesis prediction, self-driving labs.          |                 |
| <b>Modern Wavefunction Based Methods in Electronic Structure Theory</b> (Pisa, Italy)         | 27/8 – 2/9/2023 |
| Theory and implementation of quantum chemistry methods.                                       |                 |
| <b>Autumn School of Quantum Chemistry</b> (Prague, Czechia)                                   | 3 – 7/10/2022   |
| Hands-on implementation of HF, CI and MP2 on top of Psi4 (Python).                            |                 |
| <b>Swiss Equivariant Learning Workshop</b> (EPFL, Lausanne, Switzerland)                      | 11 – 14/7/2022  |
| <b>Excited States and Nonadiabatic Dynamics CyberTraining Workshop</b> (remote)               | 14 – 26/6/2021  |
| Nonadiabatic molecular dynamics methods; concluded with a small project and its presentation. |                 |

## TEACHING

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| <b>Quantum Chemistry course</b> (Dept. of Physical Chemistry, Charles University)           | 2024 – 2025   |
| DFT, HF, post-HF, semiempirical and ML approaches — theory and implementation from scratch. |               |
| <b>Hands-on instructor, Molecular Excited States Workshop</b> (Toruń, Poland)               | 9 – 13/6/2025 |
| Invited to give a hands-on tutorial on fewest-switches surface hopping (FSSH) in MLatom.    |               |

## PUBLIC OUTREACH & SCIENCE COMMUNICATION

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| <b>Open Science</b> (Supervised research internships of 8 high-school students)          | 2024–6       |
| <b>ChemQuest</b> (Evaluated high-school students' research projects)                     | 2024–6       |
| <b>Ask a Scientist</b> (Answering questions sent in by the public, published online)     | since 5/2023 |
| <b>Academix</b> (Essay for a general audience on potential energy surfaces in chemistry) | 2025         |
| <b>AI in Context</b> (Public talk on applications of machine learning in chemistry)      | 26/10/2023   |
| <b>Science Fair</b> (Presenting the research of the group/department to the public)      | 2023–5       |
| <b>FameLab</b> (Presentation and masterclass training in science communication)          | 9/2022       |
| <b>Falling Walls Lab</b> (Short research pitch)                                          | 13/9/2022    |

## REFERENCES

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|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Jiří Pittner</b>   | <a href="mailto:jiri.pittner@heu.cas.cz">jiri.pittner@heu.cas.cz</a> — PhD supervisor; Heyrovský Institute, Czech Academy of Sciences    |
| <b>Pavlo O. Dral</b>  | <a href="mailto:dral@xmu.edu.cn">dral@xmu.edu.cn</a> — Xiamen University / Nicolaus Copernicus University / Aitomistic                   |
| <b>Mario Barbatti</b> | <a href="mailto:mario.barbatti@univ-amu.fr">mario.barbatti@univ-amu.fr</a> — Aix-Marseille Université / Institut Universitaire de France |
| <b>Hans Lischka</b>   | <a href="mailto:hans.lischka@univie.ac.at">hans.lischka@univie.ac.at</a> — Texas Tech University                                         |